

CHRISTIAN RODRIGUEZ

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AREAS OF EXPERTISE

I am a Computer Science undergraduate concentrating in Information Systems at California State University, Bakersfield, with a focus on backend web development and database design. My experience includes designing relational MySQL schemas, building PHP services for role-based web portals, integrating third-party APIs for transactional email and geocoding, and deploying and administering Linux web servers. As Backend Developer on the CAPK 2-1-1 Community Resource Directory, a senior capstone project for the Community Action Partnership of Kern, I designed the database and built core backend services for a public resource directory that is replacing the agency's legacy iCarol listing system. My coursework adds systems programming in C and C++ across operating systems, networking, and parallel computing, as well as Linux administration. I also have undergraduate research experience in deploying local large language models and in statistical analysis with R and Shiny, including a peer-reviewed publication. I am seeking internships or entry-level positions in web development, data analytics, or artificial intelligence.

EDUCATION

California State University, Bakersfield

Expected Fall 2026

Bachelor of Science in Computer Science — Information Systems Concentration

GPA: 3.82

Relevant Coursework

Programming: Data Structures, Algorithm Analysis, Programming Languages

Systems: Operating Systems, Computer Networks, Distributed and Parallel Computation, Linux Administration

Web and Data: Client-Side Web, Server-Side Web, Database Systems, Statistical Computing with R

Software and AI: Software Engineering, Artificial Intelligence, Senior Project I and II

TECHNICAL SKILLS

Languages: PHP, JavaScript, SQL, Python, R, C++, C, Bash, HTML, CSS

Also Used in Coursework: C#, Prolog, Ada, Fortran

Web and Backend: PHP 8, REST/JSON APIs, session-based authentication, Leaflet.js, responsive UI design

Databases: MySQL 8 (schema design, querying, administration)

Infrastructure: Linux (Debian administration, Bash scripting, systemd, cron), Apache, DigitalOcean, Let's Encrypt SSL, DNS email authentication (DKIM, DMARC)

Systems: POSIX processes, threads, and signals; TCP/IP sockets; OpenMP

AI and Machine Learning: Hugging Face Transformers, PyTorch, LangChain, ChromaDB, retrieval-augmented generation (RAG), Gradio

Data Analysis: R (Shiny, dplyr, tidyr, ggplot2), SAS, Excel

Version Control: Git, GitHub, GitLab

SOFTWARE DEVELOPMENT EXPERIENCE

Backend Developer, CAPK 2-1-1 Community Resource Directory

Spring 2026 – Present

Senior Capstone, California State University, Bakersfield, with the Community Action Partnership of Kern

A five-person team building a public, map-based directory that helps Kern County residents find food, housing, health, and utility services, with separate portals for agency staff and CAPK administrators. Built with PHP 8, MySQL 8, JavaScript, and Leaflet.js.

- Designed the 17-table normalized MySQL schema covering agencies, programs, locations, weekly hours, service areas, languages, categories, staff accounts, invitations, reviews, and usage statistics, including a privacy flag that hides the addresses of confidential sites such as domestic violence shelters.
- Set up and administer the production LAMP server on DigitalOcean (Linux, Apache, PHP, MySQL) with Let's Encrypt SSL, including database users, permissions, and Git-based deployment.
- Integrated the Brevo transactional email API into the account invitation and password reset flows using PHP cURL, and built branded, mobile-friendly templates for both emails.

- Configured the domain's email infrastructure, with DKIM and DMARC records for outbound mail and ImprovMX forwarding for inbound support mail, after DigitalOcean's block on outbound port 25 ruled out PHP's built-in mail function.
- Migrated program listings from the legacy iCarol system into the new schema, normalizing inconsistent source records and documenting data-quality issues for the team to resolve before import.
- Wrote database seed scripts that create accounts through the application's real registration code, so password hashing and validation match production, and that resolve foreign keys at run time instead of hard-coding IDs.
- Developing automatic address geocoding for new program locations, using the U.S. Census Geocoder with a rate-limited OpenStreetMap Nominatim fallback, so listings appear on the map without paid services.
- Built the initial public directory prototype: an interactive Leaflet.js map with keyword search, location-based distance sorting, and a mobile bottom-sheet layout designed for WCAG 2.1 AA accessibility.

RESEARCH EXPERIENCE

Undergraduate Researcher, Chevron Summer Undergraduate Research Experience (SURE)

Summer 2025

California State University, Bakersfield — Mentor: Dr. Anjana Yatawara

Built a retrainable campus assistant that runs on local hardware, answering questions from the CSUB University Handbook and MATH 2200 course materials without external services.

- Fine-tuned and compared two locally deployable large language models, Mistral 7B-Instruct v0.3 for resource efficiency and DeepSeek 7B for context-specific accuracy, on the handbook and course datasets.
- Implemented a retrieval-augmented generation (RAG) pipeline with sentence-transformers, ChromaDB, and LangChain. The ingest.py script parses PDF, Word, CSV, JSON Lines, and text files, splits them with RecursiveCharacterTextSplitter, and indexes CUDA-accelerated all-mpnet-base-v2 embeddings, so adding a document updates answers without retraining.
- Developed a browser-based Gradio chat interface (chat.py) and automated environment setup, dependency installation, ingestion, and launch with PowerShell and Python scripts.
- Trained and tested on a local NVIDIA RTX A4000 workstation, the CSU NRP Nautilus GPU cluster through JupyterHub, and Google Colab, and evaluated each model on response accuracy, latency, GPU efficiency, and ease of updating its context.
- Presented the project and its results at the SURE poster session in June 2025.

Research Assistant, Air Pollution Exposure Disparities

Fall 2024 – Spring 2025

California State University, Bakersfield — Mentor: Dr. Anjana Yatawara

- Wrote R code (dplyr, tidyr, lubridate) to combine EPA daily PM2.5 monitoring files, measure missing data by county, exclude counties with more than 20 percent missing values, and compute weekly and semi-monthly county means.
- Joined pollution summaries with county median income and population data to compare exposure across income groups, and built an R Shiny app to explore PM2.5 levels by income group.
- Co-authored the resulting peer-reviewed article in Environmental Research: Health (see Publications).

ADDITIONAL PROJECTS

- *Full-Stack Web Applications* — Built web applications with PHP, MySQL, JavaScript, HTML, and CSS, including a car-detailing business site with MySQL-backed appointment booking, account registration with hashed passwords, and a PHP contact form.
- *Interactive Data Analysis App (R Shiny)* — Built a Shiny dashboard on the Palmer Penguins data with exploratory plots, regression, random forest, and gradient boosting models with cross-validation, and PCA with k-means clustering. Deployed to shinyapps.io as the STAT 3210 final project.
- *Portfolio Site* — Built <https://christianrodriguez46.github.io/>, an accessible portfolio that presents more than 100 labs and projects with documentation and live demos, using HTML, CSS, and JavaScript on GitHub Pages.

PUBLICATIONS

- Yatawara, A., Ko, K., & **Rodriguez, C.** (2025). Income-based exposure disparities across California counties, 2000–2023: A generalizable statistical framework. *Environmental Research: Health*, DOI: [10.1088/2752-5309/ae4134](https://doi.org/10.1088/2752-5309/ae4134). Introduces a reproducible, multi-pollutant, county-level statistical framework with openly shared code and data.